

Anwoy Barua

Chittagong, Bangladesh

+8801629588352 | anwoybarua@gmail.com | [linkedin.com/in/anwoy-barua/](https://www.linkedin.com/in/anwoy-barua/)

EDUCATION

Master of Science in Statistics

University of Chittagong

July 2025- June 2026

CGPA - Pending

Bachelor of Science in Statistics

University of Chittagong

Jan. 2020 – May 2025

CGPA **3.63/4** (Top 5%)

RESEARCH INTERESTS

- Climate extremes, climate risk assessment, and impact modeling
- Climate model evaluation, bias correction, and uncertainty quantification
- Machine learning and explainable AI applications in climate science
- Remote sensing and geospatial methods for climate and environmental analysis

PUBLICATIONS AND MANUSCRIPTS

Manuscripts

- Barua, A., and Ghosh, S. “Evaluation of Traditional and Machine Learning-Based Bias Correction Techniques for CMIP6 Daily Air Temperature Over Bangladesh.”** *Manuscript under preparation (Master’s thesis research).*
LightGBM demonstrated superior overall performance and improved correction of spatial and extreme temperature compared to Quantile Mapping methods.
- Barua, A., and Ghosh, S. “Validation of ERA5 and ERA5-Land Reanalysis Data for Daily Air Temperature and Extreme Indices over Bangladesh.”** *Manuscript under preparation (Master’s thesis research).*
Demonstrated that ERA5 generally provided better representation of daily temperature and extreme magnitudes, whereas ERA5-Land showed comparable performance for long-term percentile-based temperature extremes.

TECHNICAL SKILLS

- Programming:** Python (xarray, pandas, numpy, matplotlib, seaborn, scikit-learn), and R (ggplot2, dplyr)
- Tools:** ArcGIS (ArcMap10.5), Google Earth Pro, Microsoft Excel, SPSS
- Data Skills:** Statistical analysis and visualization, Machine Learning, Geospatial analysis, Remote sensing fundamentals
- Database:** PostgreSQL

LINGUISTIC PROFICIENCY

Bengali: Native

IELTS Academic: Overall **7.5** (Listening - 8.5, Reading - 7.5, Writing - 7, Speaking - 6.5)

[Certificate Link](#)

AWARDS

Information Technology Fellowship for Master’s Thesis

*Awarded by **ICT Division**, Government of Bangladesh*

Apr. 2026- Sep. 2026

PROFESSIONAL TRAINING

Mastering GIS with ArcGIS: from Fundamentals to Real-World Projects

Aug. 2025 - Dec. 2025

by *Data Solution 360*

[Certificate Link](#)

- ◇ Hands-on training in spatial analysis, geoprocessing, and remote sensing fundamentals
- ◇ Link to some of the maps created during this bootcamp. [Project link](#)

Digital Diversity, Equity, and Inclusion (DDEI) Mentorship Programme

Oct. 2023- Aug. 2024

by *BRAC James P Grant School of Public Health (JPSGH)*

[Certificate Link](#)

- ◇ Gained practical experience in literature review, research question formulation, and quantitative research design.

LEADERSHIP EXPERIENCE

President

Feb. 2024- Apr. 2025

Programming Club

[Linkedin Page](#)

Department of Statistics, University of Chittagong

- ◇ Organized Python and R workshops
- ◇ Mentored students in programming and data analysis

CERTIFICATIONS

- ◇ MOOC on “**Spatial Data Science: The New Frontier in Analytics**” — Esri, 2025 [Certificate Link](#)
- ◇ Short course on “**Basic Quantitative Research methodology**” — BRAC JPGSPH, 2024 [Certificate Link](#)
- ◇ **Introduction to Systematic Review and Meta Analysis in Health Research** — BRAC JPGSPH, 2024 [Certificate Link](#)

REFERENCES

1. **Sanjib Ghosh** (*Thesis Supervisor*)

Associate Professor

Department of Statistics, University of Chittagong

E-mail: sanjib.stat@cu.ac.bd

2. **Mrityika Barua, PhD**

Adjunct Assistant Professor and Technical Advisor

Centre for Science of Implementation and Scale-up

BRAC James P. Grant School of Public Health, BRAC University

E-mail: mrityika@bracu.ac.bd